NBS and Sustainable Drainage Systems (SuDS)

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NBS and Sustainable Drainage Systems (SuDS). Collection: Biodiversity in the Factory.

Strategic Context. Inefficient water management represents a critical operational and legal risk for aeronautical infrastructure. The saturation of conventional systems compromises aviation safety due to potential aquaplaning on runways and taxiways. The implementation of Nature-Based Solutions (NBS) acts as a mitigation strategy against extreme weather events, integrating ecosystem services that reduce asset vulnerability. Mara analyzes the plant's water risk mapping to identify critical accumulation points.

Scientific and Technical Framework. The hydrodynamics of Sustainable Drainage Systems (SuDS) are based on the restoration of natural ground infiltration rates. Through bioremediation processes, the substrate and native biodiversity act as active filters that retain heavy metals and degrade hydrocarbons from surface runoff. Silas and Mara supervise the measurement of saturated hydraulic conductivity in a test area, ensuring that technical parameters meet decarbonization objectives.

Field Analysis and Interferences. The integration of SuDS into aeronautical manufacturing environments requires a rigorous evaluation of pavement loads. The transit of large components and logistics machinery demands that drainage solutions do not compromise the soil's load-bearing capacity. Silas inspects a heavy maneuvering area where the replacement of impermeable pavements with high-resistance confinement cells is planned, verifying the absence of interferences with underground utility networks.

A determining factor in aeronautical safety is the risk of FOD (Foreign Object Debris). The design of drainage systems using native biodiversity must ensure that no volatile residues or aggregate shedding are generated that could be ingested by engines. Mara performs a field audit near a hangar, evaluating the stability of the granular materials proposed for the perimeter infiltration trenches.

Engineering of Solutions and Protocols. The design includes infiltration trenches and state-of-the-art hydrocarbon separators for primary treatment. These structures allow for source-controlled flow management, reducing the load on the general sewage network. Silas supervises the installation of a coalescent particle separator, ensuring that the connection with the native biodiversity system is watertight and efficient to maximize pollutant mitigation.

Maintenance of drainage assets must be executed under strict safety protocols and activity coordination. Mara and Silas review the preventive maintenance plan, which includes filter cleaning and technical vegetation inspection. Subcontractor management requires all external personnel to comply with the plant's PPE standards and procedures for work in confined spaces if necessary.

Industrial Reality vs. Myths. It is a technical error to consider that SuDS attract hazardous avian wildlife; a selection of non-attractive plant species mitigates strike risk. Regarding stability, the use of geocells ensures that permeable pavements support compression loads equivalent to concrete. Silas presents the final design protocols and the system's cross-section, validating asset integrity and long-term logistical durability.